Future & Promise

Future/Promise · Async thumbnail then compose URL · Build with Waqar

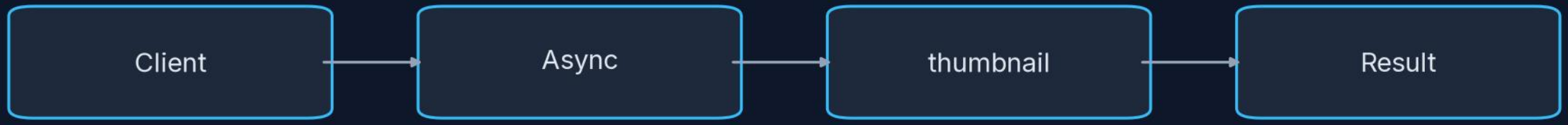
Today's Hook

- Problem: Async thumbnail then compose URL
- Pattern: Future/Promise — earned from the pain, not trivia.
- Runnable main on JDK 25 with captured stdout.
- Naive → solution → failure paths → trade-offs.

The Problem

- Async thumbnail then compose URL
- Naive design shows the cost clearly.
- Pattern isolates the force behind a clean boundary.
- Tests cover happy path and injected failures.

Structure



Future/Promise: Async thumbnail then compose URL

Real-World Scenario

- Running example: Async thumbnail then compose URL
- Same names in naive code, refactor, and Demo output.
- No shared StreamForge framing unless it fits.
- Product language first; pattern vocabulary second.

Naive Code

```java
package com.buildwithwaqar.patterns.concurrency.futurepromise;

import com.buildwithwaqar.patterns.concurrency.futurepromise.Promise;
/** Problem: generate a thumbnail asynchronously then compose the CDN URL. */
public final class UseCaseDemo {
  public static void main(String[] a) throws Exception {
    System.out.println("=== Future/Promise problem: async thumbnail ===");
    var bytes = Promise.computeAsync(128).toCompletableFuture().get();
    System.out.println("thumbBytes=" + bytes + " url=https://cdn/x/" + bytes + ".jpg");
  }
}
```

Why Naive Hurts

- Call sites couple to the wrong shape or leak on partial failure.
- Parameter drilling / duplicated envelopes / missing rollback.
- Hard to unit-test without the real dependency.
- New variants require shotgun edits.

Solution

```java
package com.buildwithwaqar.patterns.concurrency.futurepromise;

import java.util.concurrent.CompletableFuture;
import java.util.concurrent.CompletionStage;
import java.util.concurrent.Future;

/** Promise/future demo using CompletableFuture (Java's standard promise). */
public final class Promise {
    private Promise() {}

    public static CompletionStage<Integer> computeAsync(int x) {
        return CompletableFuture.supplyAsync(() -> x * 2);
    }

    public static Future<Integer> asFuture(int x) {
        return computeAsync(x).toCompletableFuture();
    }
}
```

Runnable main

```java
package com.buildwithwaqar.patterns.concurrency.futurepromise;

import com.buildwithwaqar.patterns.concurrency.futurepromise.Promise;
/** Problem: generate a thumbnail asynchronously then compose the CDN URL. */
public final class UseCaseDemo {
  public static void main(String[] a) throws Exception {
    System.out.println("=== Future/Promise problem: async thumbnail ===");
    var bytes = Promise.computeAsync(128).toCompletableFuture().get();
    System.out.println("thumbBytes=" + bytes + " url=https://cdn/x/" + bytes + ".jpg");
  }
}
```

Walkthrough Output

```
=== Future/Promise problem: async thumbnail ===
thumbBytes=256 url=https://cdn/x/256.jpg
```

Step-by-Step Trace

- Each stdout line maps to a stage above.
- Re-run the main class locally on JDK 25.
- Failure-path lines prove compensations / rejects / missing files.
- Pause and predict the next line before revealing.

Modern Java 25 / Zazr

```
// Java 25 / Zazr direction
// records + sealed + switch
// inject Env / collaborators at the edge
// virtual threads for blocking fan-out
```

Trade-offs

- Clarity vs ceremony — refuse five types for two lines.
- Teaching demos omit production metrics/idempotency keys.
- Prefer simpler language features when they erase the force.
- Know the twin pattern often confused with Future/Promise.

Related Patterns

- Contrast Future/Promise with its neighbor pattern.
- Name the roles from memory in sixty seconds.
- Point to the season map for what comes next.
- Link the GitHub episode folder in the description.

Recap

- Problem → naive → Future/Promise → Demo stdout.
- Failure paths are part of the lesson.
- Teach when NOT to use it.
- Practice: break a test, fix it, re-run the main.